

AMIR MASHALLAH ALI

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WORK EXPERIENCE

Asian Resource Centre - Intern

- Created a **Python-based** automation tool for the routine tasks i.e. categorise community member data and generate formatted reports, **increasing productivity by saving 30 minutes a day** in admin time.
- **Refined** UI/UX with new features and designs to improve accessibility and usability.
- Improved code quality and stability by fixing bugs using **React Testing Library**.
- Contributed to discussions on digital inclusion.

PROJECT AND SKILLS

Khattam

- **Developed** a mobile app with **React Native and TypeScript** on iOS to help users **log and track** missed prayers.
- **Implemented interactive calculation system** with **real-time updates** and progress tracking.
- **Created customisable tracking features** allowing users to deduct complete prayers and view total remaining instantly in real-time with **FireBase**.
- **Designed a visual dashboard** to display the progress through intuitive display boxes for easy reference.
- Built reusable UI components using **React** and **TypeScript** to enhance consistency across the codebase.

Accessibility Hackathon

- Achieved 1st place after leading a team of 8 to identify, prioritise, and propose solutions for accessibility issues of an e-commerce website which increased user retention the most amongst all participating teams.

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CERTIFICATE

Google Cybersecurity Certificate (COURSERA)

BSc Computer Science:

- **TA Finance Branch Network Design – Cisco Packet Tracer & IPv4 Networking:** I designed and fully configured a small enterprise branch network for a financial services organisation using Cisco Packet Tracer, implementing end-to-end IPv4 connectivity between departmental LANs and an external ISP. The solution included structured physical topology design using appropriate Cisco routers, switches, access points, and cabling, alongside detailed IP subnetting to support multiple departments with differing host requirements. I configured VLANs to segment Admin, Accounting, and Sales networks, implemented DHCP for dynamic address allocation, and assigned static addressing for critical infrastructure such as DNS servers. I also configured routing protocols, device security credentials, and WAN serial links to an ISP and cloud endpoint, then validated full connectivity using diagnostic tools including ping, traceroute, and interface inspection commands. This project demonstrated strong understanding of network design, IP addressing, VLAN segmentation, routing, troubleshooting, and secure configuration aligned with real-world business requirements.
- **AI Safety Gridworlds Simulation – Artificial Intelligence Project:** I designed, implemented, and evaluated multiple agent-based simulations inspired by DeepMind's *AI Safety Gridworlds* research, applying core artificial intelligence principles to model specification problems in reinforcement learning. I implemented several grid-based environments and corresponding agents, including safe interruptibility, avoiding side effects, and reward gaming scenarios, using hierarchical decision-making, state reversibility checks, and environment sensing. I developed multiple agent strategies (random walk, reward-exploiting, and goal-oriented navigation), ran controlled simulations over fixed time horizons, and quantitatively compared behavioural outcomes. The project required translating academic research into executable systems, justifying design choices, analysing emergent behaviours, and critically evaluating why naïve reward functions lead to unintended optimisation, demonstrating strong analytical thinking and practical AI implementation skills.
- **Blastonbury Pop Festival Security & Line-Up Database – Oracle, MySQL, PHP & Java:** I designed and implemented a full relational database system to support security clearance and performance scheduling for a large-scale music festival. Starting from a complex real-world specification, I produced a complete ER model, resolved many-to-many relationships, and implemented a fully normalised schema in Oracle SQL with appropriate primary and foreign key constraints. I populated the database with realistic test data and developed advanced SQL components including read-only views, access control via GRANT statements, and triggers to enforce audit logic on data deletion events. I then migrated the entire dataset, including constraints, views, and triggers, from Oracle to MySQL using WAMP, ensuring cross-platform consistency. To demonstrate end-to-end integration, I built a PHP application to query and display aggregated view data and a Java application using JDBC to retrieve and display database records. This project

demonstrated strong competency in database design, SQL programming, data integrity, and full-stack backend integration across multiple technologies.

- **War in Netaverse (WIN) Strategy Game – Java Team Project:** I contributed to the design and implementation of a Java-based strategy game prototype that simulated fleet management, resource allocation, and battle resolution within a complex rule-driven domain. Working from a detailed case study and formal interface specification, I helped design user stories, use cases, and backend domain models before implementing the system using object-oriented principles, interfaces, inheritance, and polymorphism. The solution managed multiple force types (Wings, Starships, WarBirds), each with distinct activation costs, battle strengths, constraints, and lifecycle states, alongside a dynamic war-chest economy and battle resolution engine. I implemented core gameplay logic including force activation and recall, battle selection and validation, state transitions, persistence via serialization, and both command-line and graphical user interfaces. The project demonstrated my ability to translate complex requirements into a clean, modular, and extensible architecture, apply test-first development and JUnit testing, collaborate effectively using version control and Kanban planning, and deliver a maintainable system designed for future expansion.
- **Secure IoT Network Design for Manufacturing Environments (LoRaWAN):** Designed and evaluated a secure IoT monitoring network for a semiconductor manufacturing clean room, enabling reliable long-range communication between sensor nodes and a central receiver using LoRaWAN. Implemented an IoT sensor node equipped with a DHT22 temperature and humidity sensor, visual safety indicators via multi-state LED alerts, and encrypted wireless transmission to ensure environmental compliance. Developed safety-critical logic to classify operational states (safe, warning, danger, critical) based on environmental thresholds, triggering real-time local alerts and remote notifications. Secured device-to-gateway communication using XOR-based message encryption, ensuring confidentiality during transmission. Evaluated LoRaWAN against alternative IoT protocols, analysing trade-offs in range, power consumption, latency, scalability, and security for industrial environments. This project strengthened skills in IoT architecture design, embedded programming, secure communications, industrial monitoring, and protocol evaluation.
- **Autonomous Embedded Systems Design – Vision-Based Robotic Control (ISO C):** Designed and implemented an autonomous embedded control system in ISO C for the Initio robotic car, enabling it to safely follow a target vehicle using camera-based colour detection, infrared sensors, and motor control. Developed a safety-aware control architecture incorporating safe-state behaviour, obstacle detection, distance keeping, and real-time reaction to unexpected events. Conducted resource awareness analysis by measuring execution time of critical routines and formally analysing worst-case complexity using Big-O notation. Implemented multi-threaded real-time processing with synchronisation mechanisms (mutexes) to coordinate camera processing and motion control, ensuring timely responses and bounded behaviour. Evaluated resilience, fault tolerance, and system limitations, designed degraded modes of operation, and validated behaviour through structured testing and fault-tree analysis. This project strengthened skills in embedded C, real-time systems, safety analysis, concurrency, and autonomous system design.
- **Embedded Systems & Robotics Programming Project (C, Sensor & Motor Control):** Developed an embedded control application in ISO C for the Initio robotic car, programming infrared sensors and DC motor actuators to implement real-time autonomous behaviour. Implemented obstacle avoidance by processing IR sensor inputs to safely halt and resume motion, and extended the system with a line-following algorithm using floor-facing sensors. Applied control concepts such as pulse-width modulation (PWM) to regulate motor speed and account for energy, friction, and mechanical constraints. Analysed and tuned control logic to balance speed and stability, while documenting system behaviour, limitations, and potential optimisations in a concise technical report. This project strengthened skills in embedded programming, sensor integration, motor control, and applying control theory in resource-constrained environments.
- **Ethical Technology Design Project – Social Impact Application (Team Project):** Collaborated within a multidisciplinary team to design a digital application prototype addressing a real-world societal challenge in the health, environmental, or economic domain. Applied ethical design principles and a structured design methodology to ensure inclusivity, usability, and positive social impact, while considering commercial, economic, legal, and professional constraints. Contributed to defining core features, user workflows, and value proposition, ensuring the solution aligned with user needs and responsible technology practices. Presented the prototype in a formal team presentation, communicating technical and ethical decisions clearly to a non-technical audience. This project strengthened skills in ethical system design, teamwork, stakeholder-focused thinking, and applying technology responsibly for social good.
- **Critical Technology Impact Analysis (Ethics, Law, Society & Profession):** Conducted a comprehensive 3,000-word critical analysis of a significant recent computing technology trend, examining its ethical, professional, legal, and social implications. Selected the topic through a structured review of current technology developments and relevance to society, and justified this choice based on real-world impact. Evaluated key ethical challenges (e.g., bias, privacy), professional responsibilities (standards of practice, codes of conduct), legal frameworks (data protection, compliance), and social effects (equity, accessibility), drawing on high-quality academic and industry sources. Proposed evidence-based solutions for each issue, and reflected critically on the analysis process. This work demonstrates strong skills in critical thinking, structured argumentation, multi-perspective evaluation, and the ability to link technology trends with broader societal and regulatory concerns.
- **Java Application Development – Three-Tier Architecture & MVC GUI:** Designed and implemented a full Java-based application using a three-tier architecture and the Model–View–Controller (MVC) pattern to manage competition and competitor data. Built a robust object-oriented domain model using inheritance, abstraction, and polymorphism, refactoring common functionality into a superclass and extending it through specialised competitor subclasses. Implemented data handling with ArrayLists, file-based persistence using CSV/JSON, and modular reporting features producing summary statistics, rankings, and frequency analyses. Developed an interactive GUI (non drag-and-drop) allowing real-time editing, sorting, filtering, and score recalculation, with clean

separation of concerns between presentation, logic, and data layers. This project strengthened skills in Java, OOP design, architectural patterns, GUI development, and maintainable software engineering.

- **Competition Management System – Requirements Analysis & System Design Project:** Designed a complete competition management application from requirements analysis through to system architecture. Analysed a real-world scenario involving competitor registration, scoring, role-based staff access, reporting, and data persistence, translating functional and non-functional requirements into a comprehensive UML use case model. Produced a detailed design class diagram, defining core domain entities, attributes, operations, and relationships to support extensibility and maintainability. Developed a 3-tier system architecture separating presentation, business logic, and data layers, ensuring scalability and clear separation of concerns. Considered usability, access control, data validation, and persistence using non-proprietary formats (CSV/JSON), strengthening skills in object-oriented design, system modelling, and software architecture.
- **Final Year Project:** Developed a gym progress tracking mobile application integrating AI functionalities (React Native). Implementing secure user authentication and real-time data storage via Firestore, AI model integration, and leveraged machine learning algorithms to deliver personalised training insights

MSc Cybersecurity:

- **Information Security Policy & Risk Assessment Project (Healthcare Sector):** Conducted an in-depth information security governance project based on a real-world ransomware attack on a UK healthcare provider, assuming the role of Chief Information Security Officer (CISO). Researched the organisational context, threat landscape, and regulatory environment to assess risks to sensitive clinical and operational records. Performed an asset-based risk assessment and developed a comprehensive Protection of Records Policy aligned with the ISO/IEC 27000 family, addressing confidentiality, integrity, and availability (CIA) requirements. Evaluated legal, regulatory, and contractual obligations including GDPR and NHS-related compliance, and mapped appropriate technical and organisational controls to reduce residual risk to acceptable levels. This project strengthened skills in security governance, risk management, policy development, and applying international security standards to complex, high-impact environments.
- **Digital Forensics Investigation (Computer Evidence Analysis):** Conducted a full digital forensic investigation as part of a simulated Customs and border enforcement case, analysing a forensic image of a suspect's computer in accordance with evidential integrity and continuity requirements. Set up a dedicated forensic workstation and verified evidence integrity using cryptographic hash values prior to analysis. Performed a structured forensic examination of the disk image using appropriate forensic tools, documenting all actions through detailed contemporaneous notes to maintain evidential credibility. Produced a formal investigative report outlining methodology, findings, and conclusions, demonstrating strong capability in evidence handling, forensic analysis, and professional reporting suitable for legal and regulatory contexts.
- **Grey-Box Penetration Testing Project (Team-Based Cybersecurity Assessment):** Worked as part of a consultancy-style penetration testing team to deliver a full-scale grey-box penetration test against a live server environment. Contributed to planning and managing the engagement lifecycle, adapting roles and techniques as requirements evolved. Conducted in-depth vulnerability discovery, exploitation, and privilege escalation to root/admin level across multiple attack paths, confirming and documenting at least five unique high-impact attacks. Critically evaluated security controls within a complex, heterogeneous environment and produced clear, business-focused recommendations tailored for senior decision-makers. Delivered a professional penetration testing report including executive risk assessment, technical attack details, and mitigation strategies, strengthening skills in offensive security, teamwork, client communication, and translating technical risk into business impact.
- **Threat Modelling & Attack Tree Analysis (Individual Cybersecurity Project):** Completed an individual research-driven cybersecurity project focused on analysing the 2025 threat landscape, critically evaluating modern attack techniques, threat actors, and emerging risks. Designed a comprehensive Attack Tree for a fictional organisation, defining a realistic threat profile and modelling multiple attack paths aligned with current industry and academic research. Applied advanced threat modelling and penetration-testing concepts to assess vulnerabilities and evaluate the effectiveness of security controls, procedures, and defensive techniques. Delivered evidence-based recommendations mapped directly to the attack scenarios, supported by recent academic literature and industry best practices. This project strengthened skills in threat modelling, risk analysis, security architecture evaluation, and communicating strategic security advice to decision-makers.
- **Cyber Defence & Incident Response Project (PCAP Analysis & Network Hardening):** Acted as a cybersecurity analyst in a simulated national defence scenario, conducting post-incident analysis of a suspected cyberattack targeting Critical National Infrastructure (CNI) and a National Military Command and Control System (NMC2S). Performed in-depth PCAP analysis using Wireshark to identify attacker reconnaissance, exploitation attempts, and potential data exfiltration, assessing impact and risk using recognised frameworks such as NIST / ISO 27001. Analysed attacker TTPs to infer objectives and future threat vectors, contributing counterintelligence insights to strengthen defensive posture. Designed and implemented a Bash-based defensive proof of concept on Kali Linux, automating mitigation actions such as IP blocking and suspicious traffic detection. Delivered a formal technical report with evidence-backed findings, strengthening skills in incident response, threat analysis, network defence automation, and security reporting within high-stakes environments.
- **Data Analysis & Visualisation Project: Spotify Music Trends (2000s):** Collaborated as part of a group research project analysing the relationship between beats per minute (BPM) and song popularity on Spotify during the 2000s. Contributed to refining the research question, analysing datasets, and producing clear, insightful visualisations using R Studio to identify trends and correlations. Worked collaboratively to incorporate feedback from an earlier submission and deliver a structured, professionally written report within strict deadlines and word-count constraints. This project strengthened skills in data analysis, statistical

reasoning, data visualisation, and team-based project delivery, while demonstrating the ability to communicate technical findings clearly and effectively.

- **Web Accessibility Mini-Hackathon (Team Project):** Participated in an intensive, time-boxed mini-hackathon focused on identifying and prioritising web accessibility issues in a live website environment. Worked collaboratively within a multidisciplinary team to audit accessibility compliance using industry-standard tools such as WAVE, assessing alignment with global legislation including the UK Equality Act 2010, ADA, and WCAG principles. Contributed to analysing high-impact accessibility barriers affecting users with disabilities and proposed practical, standards-compliant solutions within a three-hour deadline. Produced clear, structured documentation summarising findings and recommendations, strengthening skills in teamwork, rapid problem-solving, accessibility compliance, and inclusive digital design.
- **Ethical, Legal and Professional Issues in Computing Project:** Completed an individual, time-constrained academic project focused on analysing ethical, legal and professional issues within contemporary computing scenarios. Critically evaluated high-profile case studies using recognised professional frameworks such as BCS and ACM codes of conduct, demonstrating a strong understanding of ethical decision-making, legal compliance and professional responsibility. Produced structured, well-referenced written outputs that adhered to academic, legal and ethical standards, showcasing the ability to analyse real-world technological, social and commercial implications under pressure. This project strengthened my critical thinking, professional judgement and ability to apply ethical standards to real-world computing and cybersecurity contexts.
- **Masters Dissertation:** Developed a Python-based hybrid analysis pipeline for Android applications to statically and dynamically analyse APKs to identify vulnerabilities and produce JSON file reports. Fully engaged in all stages of development during the project, from conceptualisation, to setting up and running the application through powershell, testing and comparing analyses tools such as MobSF and Frida for evaluation purposes, proposing mitigation strategies aligned with OWASP Mobile Top 10, and leveraged machine learning to classify detected issues by severity to overall improve reporting accuracy.

EDUCATION

University of Hertfordshire, BSc Computer Science, 1st Class Honours

- Elective modules: Software Engineering, Artificial Intelligence, Internet of Things, Software Architecture

University of Hertfordshire, MSc Cyber Security, 1st Class

- Elective modules: Cyber Operations, Responsible Technology, Information Security and Compliance, Penetration Testing

VOLUNTEERING

Who Is Hussain? - London

- Started as a driver (2021) and grew to coordinator and organiser (2023).
- Lead weekly food drives in Acton and Charing Cross with rotating volunteer teams
- Promoted to Head of Drivers (2025), responsible for ensuring the van's insurance and tax compliance and reporting directly to the London Lead.
- Currently Deputy Head of Drivers, overseeing vehicle readiness, training new drivers, making sure we have drivers signed up for the food drives, and maintaining consistent communication across the team.

University Society Vice-President/Event-Organiser

- Co-led society and organised events with guest speakers, and collaborated with different societies across London.
- Looked after university societies in London to ensure their needs are met.
- Attended weekly London meetings, and monthly general-team meetings to discuss ongoing support for university students and exchange updates.
- Hosted a national football event with speakers and 200+ attendees from all over the UK, raising £5400 for Al-Ayn charity for the orphans.

Vice Chair - MSC

- Overseeing 45+ societies nationally, ensuring strong engagement and communication with leaders.
- Build trusted relationships with society leaders to strengthen relationships and provide unique support for unique occasions/scenarios.
- Updating essential documents in Google Workspace to keep all committee members informed and aligned.
- Organise national summits, retreats, training days, and conferences - exceeding 200+ sign-ups for core projects through proactive outreach.
- Travel nationally to different universities to advise and support societies with internal issues, maintain strong communication points, network with different students to collaborate on events.